Qualitative Analysis of the Yamuna River Pollution

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Abstract

The purpose of this research was to determine how water quality varies throughout the Yamuna River with respect to pollution. Instead of testing multiple water quality parameters individually, we developed an integrated index called the Water Pollution Index (WPI) that includes the BOD, DO, pH, faecal coliform and nitrates values as one single value. A C++/Python regression model was employed to develop a formula for the WPI and was evaluated on both clean and polluted portions of the river. The results were in general agreement with our expectations, therefore indicating that the WPI approach is applicable. As such, the WPI greatly simplifies the ability to evaluate different locations within a river and assess changes in pollution levels over time. Consequently, the application of the WPI by researchers and government agencies will provide them with greater insight into river pollution and assist them in implementing improvements to water quality.

We have included the population data along with and plotted its graph which shows the industrialisation of the station increases with the increment of population and the graph shows that where population increases pollution increases.

1 Introduction

River pollution has become a serious issue, mostly in regions experiencing rapid industrial growth. Rivers are essential for drinking water, agriculture, and ecosystem health, despite of this they are continuously contaminated. Analysing water quality of river is often difficult because it requires analysing many parameters at the same time, which makes interpretation more difficult. To simplify this process, we introduced a single factor called the Water Pollution Index (WPI). WPI combines multiple water quality parameter like BOD,COD into one value i.e (WPI),which provides a clear indication of the river's condition. Using regression modelling in C++, we have developed a formula for WPI that can help compare pollution levels across different rivers and track changes in pollution levels of a river with time through a single equation of WPI which we have developed.

1.1 Literature Review

Many studies have examined river pollution, showing that sewage, industrial waste, and farm waste are the main reasons why rivers become polluted contaminated. Most researchers rely on several water quality parameters like pH, DO, BOD etc.. to judge a river's condition, but looking at all these values together is tough. Building on this idea, our work introduces a Water Pollution Index (WPI). Using regression modelling in Python, we created a single formula that combines multiple parameters like BOD,COD,$PO_4{}^{3-}$,NO^{3-},dissolved oxygen into one easy-to-interpret value. This helps compare different rivers more clearly and conclude how their pollution levels change over time without involving in a lot of complications.

1.2 Factors Considered

Our model simplifies the analysis of river by pollution by introducing a single term WPI(Water Pollution Index) based on regression of dataset on the following factors:-

- **BOD(Biochemical Oxygen Demand)**
  BOD measure how much oxygen microorganism need to break down organic matter in water. High BOD means there is a lot of waste or decaying material present. This reduces oxygen available for fish and aquatic life, making the river unhealthy.
- **pH**
  pH measures how acidic or alkaline the river water is. Most aquatic organisms survive only in a narrow pH range. Changes caused by industrial discharge or acid rain can harm aquatic life and disrupt chemical balance.
- **Dissolved Oxygen (DO)**
  DO tells us how much oxygen is present in the water for aquatic organisms. Low DO levels signal poor water quality and can lead to fish kills. Pollution, high temperatures, and excessive organic matter often reduce DO.
- **Faecel Coliform**
  Faecal coliform bacteria basically show that the water is dirty because of sewage or animal waste. If their level is high, then the water is not safe to drink or even to play in. These bacteria can cause many diseases, and it also means that the sanitation in that area is not proper. So overall, high FC means the river or lake is kinda polluted and not good for people.
- **Nitrate(NO^{3-})**
  Nitrates come from fertilizers, sewage, and agricultural runoff. Excess nitrates promote algal blooms, reducing oxygen in the water. They also pose health risks, especially for infants, if the water is consumed.

1.3 Objectives of the Study

The main goal of this study is to create a simplified way to analyse river pollution using a Water Pollution Index (WPI) and for that we focuses on the following objectives:

1. **To define a WPI scale ranging from 0 to 1** that clearly represents the pollution level of a river, making it easier to compare water quality across different regions of Yamuna.We do this by researching on google about pollution level of various regions of Yamuna river.We got that by pure logic that Yamuna should be the cleanest at its source.Also we found that Yamuna is most polluted around regions around Delhi Okhla. Also its Moderate Polluted around Chambal Area.
2. **To simplify the analysis of river pollution by developing a regression-based equation for WPI** using 5 water factors like BOD, DO, pH,faecal coliform, and NO^{3-} so that the overall condition of a river can be expressed through a single value.
3. **See how River Pollution varies with Factors like Industrial Growth and Population Growth** by comparing the graph for other Regions pridicted WPI.

2 Problem Formulation

Previous studies on river water quality in India, including work that analyzed eight major rivers from 2001 to 2010, relied on assigning fixed weighting factors to factors such as BOD, DO, COD, TSS, and faecal coliform. These weights are typically derived using sensitivity analysis, where a higher weight is assigned to a factor that causes a larger percentage change in overall water quality for small variations in its value. Even though these methods show which factors matter more,

BOD Level (in ppm)	Water Quality
1 - 2	**Very Good** There will not be much organic waste present in the water supply.
3 - 5	**Fair: Moderately Clean**
6 - 9	**Poor: Somewhat Polluted** Usually indicates organic matter is present and bacteria are decomposing this waste.
100 or greater	**Very Poor: Very Polluted** Contains organic waste.

Figure 1: BOD Level Classification

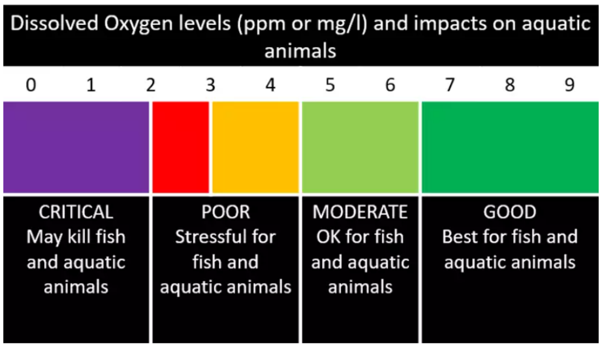

Figure 2: DO Level Classification

they have drawbacks. The weights never change for different rivers, and they also cannot predict pollution for new river locations.

In this paper, we make a regression model for the Yamuna to get one pollution score. We take the clean starting point near Yamunotri as $Y = 0$ Y=0 and the highly polluted Shahdara drain area as $Y = 1$ Y=1. All other places are given values between 0 and 1 based on news, photos, and reports. This gives a simple pollution scale made just for the Yamuna.

For every river segment, measurements of 4 factors are collected and then we formed the needed matrix.

$$X = [\mathrm{BOD}, \mathrm{DO}, \mathrm{NO}_3, \mathrm{pH}].$$

Let X be the data of m river locations and Y be their pollution values. We use Linear Regression to find the relationship between the water quality factors and the pollution level. The model predicts the pollution value using:

$$\hat{Y} = WX + b,$$

where $W = [w_1, w_2, \ldots, w_9]$ represents the regression coefficients, and b is the intercept term. On expanding this model becomes

$$\hat{Y} = b + w_1(\mathrm{BOD}) + w_2(\mathrm{DO}) + w_3(\ \mathrm{pH}) + w_4(\mathrm{NO}^{3-})$$

The goal of the regression model is to find the best values of W and b that match the real pollution values Y. We do this by minimizing the Mean Squared Error (MSE).

$$J(W, b) = \frac{1}{m} \sum_{i=1}^{m} \left(\hat{Y}^{(i)} - Y^{(i)} \right)^2.$$

The optimization problem is therefore

$$\min_{W,b} J(W, b).$$

Taking derivatives yields the gradients used in parameter estimation:

$$\frac{\partial J}{\partial W} = \frac{2}{m} X^T (WX+b-Y), \qquad \frac{\partial J}{\partial b} = \frac{2}{m} \sum_{i=1}^{m} (\hat{Y}^{(i)} - Y^{(i)}).$$

Using Ordinary Least Squares (OLS) or Gradient Descent, we find the best coefficients. This gives us a final equation for the Yamuna's Water Pollution Index (WPI

$$WPI(x) = w_1 x_1 + w_2 x_2 + \cdots + w_5 x_5 + b,$$

This equation can predict the pollution level of any new Yamuna location using its water quality data. Unlike older methods with fixed weights, our equation is fully based on real data. Positive values mean a factor increases pollution, while negative values mean it improves water quality. This makes the model simple, useful, and well-suited for studying pollution in the Yamuna.

3 Numerical Analysis

In this study, we examine how WPI is related to different water quality factors along the Yamuna. We use Multilinear Regression because the data shows that pollution changes almost linearly with these factors.

The regression model used in this work is expressed as:

$$Y = b + w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4$$

where Y represents the assigned pollution potential at a given river segment, and the variables x_1 to x_4 correspond to the five water quality parameters: BOD, DO, $\mathrm{NO}_3$and pH.

TThe regression coefficients were calculated using the matrix least squares method, which gives the best fit by minimizing the error. This helps reduce the difference between the real pollution values and the values predicted by the model.

Using this method helps us see how each water quality factor affects the Yamuna's pollution. The final linear model is easy to understand and gives clear, data-based insights without adding extra complexity

3.1 Matrix Least Square Method Derivation

3.1.1 Linear Regression Derivation

Consider a simple linear regression model of the form

$$y = wx + b,$$

where x is the input variable, y is the output, w is the slope, and b is the intercept. For m observations, the model can be written in vector form as

$$Y = X\phi,$$

where

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix}, \quad \phi = \begin{bmatrix} b \\ w \end{bmatrix}, \quad Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}.$$

The goal is to find parameters ϕ that minimize the squared error:

$$J(\phi) = \|Y - X\phi\|^2.$$

Expanding,

$$J(\phi) = (Y - X\phi)^T (Y - X\phi).$$

Taking the derivative of $J(\phi)$ with respect to ϕ:

$$\frac{\partial J}{\partial \phi} = -2X^T Y + 2X^T X\phi.$$

Setting the derivative equal to zero:

$$X^T X\phi = X^T Y.$$

Solving for ϕ gives the well-known Normal Equation:

$$\phi = (X^T X)^{-1} X^T Y.$$

This solution provides the optimal linear regression coefficients under the least squares criterion.

3.1.2 Multilinear Regression Derivation

For Multilinear Regression with n predictors, the model generalizes to:

$$Y = X\phi,$$

where

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1n} \\ 1 & x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix}, \phi = \begin{bmatrix} b \\ w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}, Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}.$$

The error minimization follows exactly the same procedure as in simple linear regression. The cost function is

$$J(\phi) = \|Y - X\phi\|^2,$$

and differentiating with respect to ϕ again yields

$$X^T X\phi = X^T Y.$$

Thus, the solution for Multilinear Regression remains identical to the linear case:

$$\phi = (X^T X)^{-1} X^T Y. \tag{1}$$

This formula provides the optimal coefficients for any linear or multilinear regression model under the least squares approach.

4 Results and Discussion

We developed C++ codes for Linear Regression, Multilinear Regression Methods.Google Sheets was used to visualize the variation of water quality parameters along different locations of the Yamuna River . To understand how the river's water quality changes from the source to the downstream areas, we made graphs for BOD, COD, faecal coliform, nitrate, and pH. These graphs helped us see how each factor increases or decreases as the river moves through different places.

After that, we checked how each of these factors is linked to the Water Pollution Index (WPI). We looked at all of them together in multivariate plots and also compared them one by one. From this study, we noticed that most of the data followed a straight-line pattern. Because of this clear trend, we decided that Multilinear Regression was a good method to model how pollution changes along the river.

4.1 Prediction Models Developed

Below, we give the final equation that we will use to predict the WPI. To check if this equation works well, we will test it using data from one place near the river's clean starting point and another place from a very polluted area We also check how the pollution changes when we go from north part of Yamuna to south part. This show us how the river get more dirty when more factories and people are near it.

Here is the equation of WPI which we get after using regression on the dataset Yamuna River

$$\begin{aligned} \text{WPI} = {} & (-0.0296585 \times \text{DO}) \\ & + (0.340454 \times \text{pH}) \\ & + (0.0104573 \times \text{BOD}) \\ & + (0.137182 \times \text{NO}^{3-}) \\ & + (-2.15888) \end{aligned} \tag{2}$$

4.1.1 Verification of our WPI equation

Now we will verify our WPI equation by taking the cleanest part the Yamuna River(Hanumanchatti) and the most polluted part of the Yamuna River(Nizamuddin)

For Hanumachatti
DO_{avg}=9.7 pH_{avg}=7.2
BOD_{avg}=1 NO^{3-}_{avg}=0.33
after putting all these values in our WPI equation(Written above we get the value of WPI)
WPI=0.06042871

For Nizammudin part of Yamuna River
DO_{avg}=1.75 pH_{avg}=7.05
BOD_{avg}=30 NO^{3-}_{avg}=0.9
after putting all the values in our WPI equation we get the value of WPI
WPI=0.830873525
In the above two examples we get the value of

WPI close to 0 for the cleanest part of Yamnuna river and close to 1 hence we can conclude that our equation is working correctly.

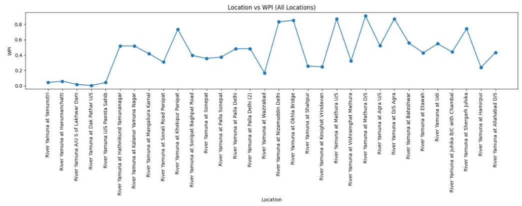

Figure 3: WPI vs Location

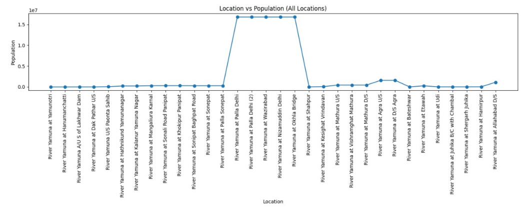

Figure 4: WPI vs Population

4.2 WPI Trends

After proper analysis the following trends and relations were observed.

4.2.1 BOD

Higher BOD make the river more dirty because more oxygen is need to break the waste in water. When BOD go up, dissolved oxygen go down, and fish and other animals find it very hard to live. This also make water bad and sometimes it smell very bad. More harmful germs and bacteria start to grow in the water. All these things together show that the river is under strong pollution and the water is not safe for life.

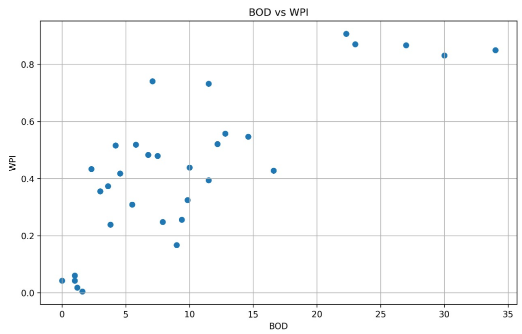

Figure 5: WPI vs BOD

4.2.2 Disolved Oxygen

Lower DO make river more polluted because there is not enough oxygen for fish and other animals in water to live. Pollutants like waste and chemicals use oxygen when they break down, and this make DO go down. When DO get too low, river become weak and sick. Fish, plants, and other animals cannot live normal in it. The water become unhealthy, sometimes it smell bad, and more harmful germs can grow. All this show that river is in trouble and the pollution is very high.

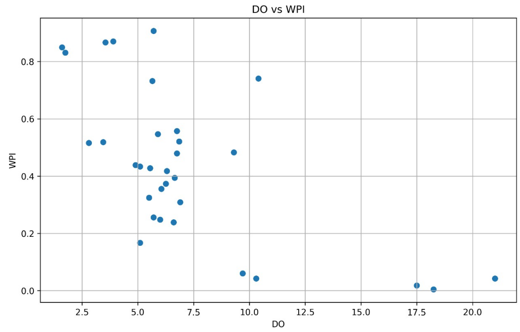

Figure 6: WPI vs DO

4.2.3 pH

Unbalanced pH make river more polluted because most fish and animals in water can live only in small pH range. If water become too acidic or too alkaline, it usually happen because of factories, sewage, or chemicals coming from land. This make problems for animals and plants in water and also disturb the natural balance of river. The river become unhealthy and some animals cannot survive in it.

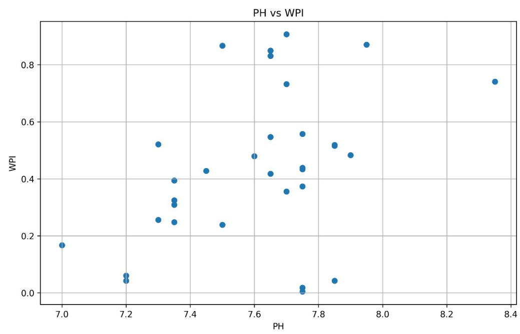

Figure 7: WPI vs pH

4.2.4 Nitrate

High nitrate levels increase pollution because nitrates from fertilizers, sewage, and runoff promote excessive algal growth. This leads to eutrophication, where algae consume oxygen and cause DO to drop. As oxygen levels fall, fish die, and the river becomes heavily degraded.

5 Conclusion

At the end of our study, we mainly tried to see how the pollution levels change in different parts of the Yamuna River. Instead of looking at many separate test values,

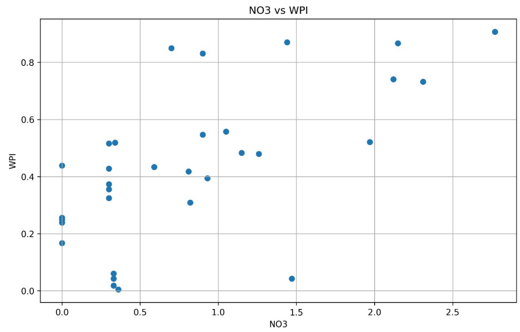

Figure 8: WPI vs NO^{3-}

we made one simple number, the Water Pollution Index (WPI), which honestly made the whole thing much easier to understand. We used data from different places and made a regression formula in C++/Python, and when we checked it on clean and polluted spots, the results were almost what we expected, so the formula seems to work fine.

From the graphs and the data, it is clear how things like BOD, DO, pH and nitrates affect the river. Some areas were way more polluted, especially where population and industries are high. With WPI, comparing these places became much simpler because everything comes down to just one number instead of a bunch of confusing values.

Overall, WPI turned out to be a useful and simple way to understand river pollution. Even someone who doesn't know much about water quality can look at the WPI and get an idea of how clean or dirty the river is.

6 Self-Assessment

Level 1

We did level 0 as we did complete literature assessment of the formulae and prediction models used in our referred research papers. We followed multi linear regression to find WPI as a function of DO, BOD, pH, NO^{3-} which gave us linear equations.

Level 2

First we have extracted the data of Yamuna River of the factors like BOD, pH, NO^{3-},DO and then used multilinear regression to derive the equation of WPI. We have developed programmes for Multilinear regression in 4 independent variables for calculating our WPI equation. Our code is perfectly working as we have provided.

Level 3

We have cited some of the research papers and some websites which provided the data of river pollution in many parameters separately like that of chemical concentrations, Fecal Coli etc. and that too in a complicated way. So, we have used C++ coding, developed the code for multi linear regression to solve this problem by developing a single factor for assessing the water pollution in a river. Due to this factor even a lame man can understand the level of pollution in a river. In short we have made a term similar to AQI for rivers.

7 References

1. P. Chawla, P. Kumar, M. Singh, N. Hasteer and C. Ghanshyam, "Prediction of pollution potential of Indian rivers using empirical equation consisting of water quality parameters"

2. 2015 IEEE Technological Innovation in ICT for Agriculture and Rural Development (TIAR), Chennai, India, 2015, pp. 214-219, doi: 10.1109 /TIAR.2015.7358560.

3. Water Quality Data of Rivers Monitored Under National Water Quality Monitoring Programme (NWMP), 2021